MATH290

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Sep 21, 2026

Make and prove a conjecture about a common divisor shared by all integers of the form $7^n - 4$ for n a non-negative integer.

Proposition 1. *All integers of the form $7^n - 4$ for a non-negative integer n are divisible by 3.*

Proof. The proof is by induction on n. Consider a base case of $n = 0$. Observe,

$$7^0 - 4 = 1 - 4 = -3$$

-3 is divisible by 3, so the result holds. Now suppose that the result holds for some non-negative integer k. That is, $7^k - 4$ can be expressed as the product of 3 and some other integer j. Observe how we can rearrange this identity:

$$\begin{aligned} 7^k - 4 &= 3j \\ 7^k &= 3j + 4 \end{aligned}$$

Now consider the case for $k + 1$. Observe,

$$\begin{aligned} & 7^{k+1} - 4 \\ &= 7(7^k) - 4 \\ &= 7(3j + 4) - 4 \quad \text{[by the inductive hypothesis]} \\ &= 21j + 28 - 4 \\ &= 21j + 24 \\ &= 3(7j + 8) \end{aligned}$$

$7j + 8$ is itself an integer. Thus, the result holds for the inductive case. Therefore, by the principle of mathematical induction, the result holds for any integer $k \geq 0$.

□